MOHD ABDUL FAIZ

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Mechanical Engineering Undergraduate / Automotive & Design Focused

PROFESSIONAL SUMMARY

Mechanical Engineering undergraduate at CBIT Hyderabad with hands-on expertise in SolidWorks, ANSYS Fluent, and CFD/FEA simulation. Experienced in Formula SAE-style chassis design, steering geometry, geometry optimisation, thermal & structural analysis, and emission control systems. Proficient in Python and data science. Two-time award winner with active leadership in national-level technical events. Seeking an internship or entry-level role in high-performance, sustainable engineering.

EDUCATION

Bachelor of Engineering in Mechanical Engineering Aug 2023 – May 2027

Chaitanya Bharathi Institute of Technology (CBIT)(A), Hyderabad, India

- Core Subjects: Thermodynamics, Fluid Mechanics, Machine Design, Manufacturing Processes, CAD/CAM, Heat Transfer
- Member :– SAE Club CBIT (Society of Automotive Engineers) & ASME (American Society of Mechanical Engineers)
- Core Organiser – Mechanica 2026, National Level Technical Fest; Total Documentation Head and lead the Events Teams

TECHNICAL SKILLS

CAD Software: SolidWorks, AutoCAD, Fusion 360, etc

CAE / Simulation: ANSYS Fluent, ANSYS Mechanical, ANSYS Workbench, CFD-Post

Analysis Types: CFD, FEA, Thermal Analysis, Structural Analysis, Flow Simulation

Engineering Core: Thermodynamics, Fluid Mechanics, Machine Design, CAD/CAM, Manufacturing

Programming: Python (Advanced – 100 Days of Code Bootcamp), Data Science Fundamentals

Soft Skills: Research & Problem Solving, Analytical Thinking, Technical Writing, Mentorship, Team Management

EXPERIENCE

Mechanical Engineering Intern | Prahethi Racing — Hyderabad, India

Technical Design & Analysis — Chassis Development

- Designed a Formula 5 car chassis in SolidWorks from initial concept to full structural validation; ensured FSAE compliance throughout the design lifecycle
- Applied FEA principles, including basic meshing to verify structural viability and design correctness of critical chassis components
- Engineered Ackermann steering geometry integrated with Rear-Wheel Steering (RWS), improving vehicle handling dynamics and cornering precision

Materials & Component Engineering

- Audited fasteners from prior vehicle iterations; researched Titanium and Alloy Steel to achieve optimized weight-to-strength ratios
- technical specifications and governing standards for fasteners to ensure trackside safety and technical inspection compliance

Ergonomics & Regulatory Compliance

- Implemented ergonomic cockpit layout procedures optimised for driver performance and competition requirements
- Interpreted FSAE regulations on seat and firewall construction, ensuring full vehicle safety compliance

General Contributions

- Modelled and validated vehicle sub-systems in SolidWorks; collaborated with cross-functional teams on fabrication, testing, and performance benchmarking
- Gained exposure to material selection, manufacturing tolerances, and vehicle dynamics in a competitive motorsport environment

PROJECTS

Geometry Optimization for Flow & Performance in Catalytic Converters | SolidWorks, ANSYS Fluent

- Parametrically modelled multiple catalytic converter geometries varying substrate shape, cone half-angle (10°–30°), and inlet/outlet configurations to study exhaust gas flow behaviour
- Executed CFD simulations using the k-ε turbulence model; analysed velocity contours, pressure drop, flow uniformity index (UI), and wall shear stress across all variants
- Identified optimal geometry with highest flow uniformity and minimum pressure drop, improving catalytic efficiency and reducing harmful exhaust emissions
- Validated results against fluid mechanics and emission control theory; documented findings in a technical report with comparative tables and contour plots

NACA 4412 Airfoil: CFD Analysis & Validation | SolidWorks, ANSYS Fluent

- Modelled the NACA 4412 airfoil in SolidWorks and configured structured CFD simulations toanalysee C_L, C_D, pressure distribution, and velocity profiles across angles of attack
- Validated simulation results against published experimental data; demonstrated strong integration of aerodynamics theory and mesh generation strategies

Case Studies in Mechanical Systems and Design of Formula 5 Car | SolidWorks, AutoCAD

- Conducted case studies on mechanical component design, load analysis, and failure mode evaluation; proposed and analytically validated design improvements

CERTIFICATIONS & TRAINING

- **Computational Fluid Dynamics (CFD) Course** – Mesh generation, solver setup, turbulence modelling, and post-processing
- **100 Days of Code: Python Pro Bootcamp** – Advanced Python programming, automation, and data handling
- **Data Science Foundation – Springboard** – Statistics, data wrangling, exploratory analysis, and visualization
- **National Level 3D Printer Workshop** – Additive manufacturing, FDM printing, and Design for AM
- **Workshop: Advances in Design, Manufacturing & Testing of Composites** – CBIT Dept. of Mechanical Engineering in collaboration with ASL, DRDO
- **Department of Mechanical Engineering Bootcamps – CBIT** – Industry-oriented technical skill development sessions

ACHIEVEMENTS & AWARDS

Winner – Inter-College Paper Presentation & Poster Presentation Competition
Chaitanya Bharathi Institute of Technology

- Secured 1st place in a competitive inter-college technical event; presented original research to a faculty panel, recognised for technical depth and communication

Runner-Up – Mechanical Engineering Hackathon

- Designed and prototyped an innovative engineering solution under time constraints; ranked among top teams for technical creativity and execution

AREAS OF INTEREST

CFD · FEA · Automotive & Motorsport Engineering · Emission Control Technology · Sustainable Energy Systems · Product Design & Optimization · Additive Manufacturing · Composite Materials · Data-Driven Engineering